Two coordinate distributions → one shared MACE architecture → three model routes

MigrationBench training designs: broad pretraining coverage and compact task-specific coverage

Coordinate distributions

Descriptors

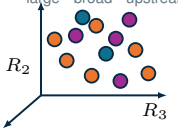
Shared MACE architecture

Training routes

Resulting models

Foundation / pretraining data

large • broad • upstream



same descriptor map

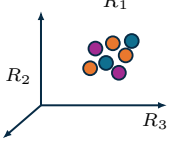
pretraining descriptors

$$\mathbf{G}^{\mathrm{pre}} = [\mathbf{G}_1, \ldots, \mathbf{G}_N]$$

$$\mathbf{G}_i^{\mathrm{pre}} = [G_i^1, G_i^2, G_i^3, \ldots]$$

In-house migration data

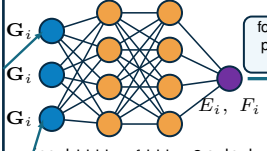
small • curated • target-specific



in-house descriptors

$$\mathbf{G}^{\mathrm{in}} = [\mathbf{G}_1, \ldots, \mathbf{G}_m]$$

$$\mathbf{G}_i^{\mathrm{in}} = [G_i^1, G_i^2, G_i^3, \ldots]$$



data / checkpoint provenance is explicit

foundation / pretraining data → PRETRAIN on broad data → Foundation / Generalist model use as-is / no update

route-specific update

in-house migration data → TRAIN FROM SCRATCH → Specialist model in-house only

foundation checkpoint + in-house migration data → FINE-TUNE on in-house data → Fine-tuned model foundation init; in-house update

two inputs are required

Qualitative coordinate-distribution sketches; not to scale. Colors distinguish data, training/update steps, and model/checkpoint lineage.